

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Flexential - Richmond, VA -- station KOFP, America/New_York
Building OSM 257581364
Flexential - Richmond
Facade gap not applicable -- one building, no second facade
Weather record 43,547 real hourly records from KOFP
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-02-09 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 22 hours with 1 mode change. The reactive incumbent took 12 hours -- 4 more -- but declared 2 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 8 of 22 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 22 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
01	mechanical	yes	switch budget	-2.505	18.0	-2.780	0.897
02	mechanical	yes	switch budget	-2.500	18.0	-2.780	0.811
03	mechanical	yes	switch budget	-2.500	18.0	-2.780	0.781
04	mechanical	yes	switch budget	-2.750	18.0	-3.280	0.748
05	mechanical	yes	switch budget	-2.750	18.0	-3.280	0.722
06	mechanical	yes	switch budget	-2.475	18.0	-3.280	0.817

07	mechanical	yes	switch budget	-1.365	18.0	-2.780	1.266
08	mechanical	yes	switch budget	4.745	18.0	6.110	1.781
09	mechanical	yes	switch budget	9.470	18.0	13.330	1.991
10	mechanical	yes	switch budget	13.890	18.0	21.110	2.062
11	mechanical	no	dry-bulb	18.060	18.0	22.780	1.401
13	mechanical	yes	-	16.390	18.0	15.000	0.916
14	mechanical	no	dry-bulb	19.995	18.0	15.000	1.058
15	mechanical	no	dry-bulb	20.280	18.0	15.000	0.960
16	FREE	yes	-	15.280	18.0	13.890	1.011
17	FREE	yes	-	13.335	18.0	11.110	1.142
18	FREE	yes	-	12.220	18.0	9.440	1.501
19	FREE	yes	-	10.275	18.0	7.220	1.630
20	FREE	yes	-	8.885	18.0	7.220	1.655
21	FREE	yes	-	8.055	18.0	6.670	1.611
22	FREE	yes	-	6.945	18.0	6.670	1.177
23	FREE	yes	-	6.945	18.0	6.670	0.867

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.505 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.750 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.750 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.475 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.365 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.745 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.470 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.060 C against a 18.0 C limit, so it fails by 0.060 C. It would take a limit 0.060 C higher, or a bound 0.060 C tighter, to change this hour.

13:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.390 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.995 C against a 18.0 C limit, so it fails by 1.995 C. It would take a limit 1.995 C higher, or a bound 1.995 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.280 C against a 18.0 C limit, so it fails by 2.280 C. It would take a limit 2.280 C higher, or a bound 2.280 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.280 C, which is 2.720 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.011 C, and

the margin is measured, not chosen: 1.011 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.335 C, which is 4.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.142 C, and the margin is measured, not chosen: 1.142 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.220 C, which is 5.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.501 C, and the margin is measured, not chosen: 1.501 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.275 C, which is 7.725 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.630 C, and the margin is measured, not chosen: 1.630 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.885 C, which is 9.115 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.655 C, and the margin is measured, not chosen: 1.655 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.055 C, which is 9.945 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.611 C, and the margin is measured, not chosen: 1.611 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.945 C, which is 11.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.177 C, and the margin is measured, not chosen: 1.177 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.945 C, which is 11.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.867 C, and the margin is measured, not chosen: 0.867 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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